

[CS112] Links and Resources

Source: Official Course Pages and Reviews

1. Fall 2025 Course Catalog Page:

- Link: <https://www.cs.tufts.edu/t/courses/description/fall2025/CS/112-01>
- Contains official course description and registration information

2. Instructor Rating and Reviews:

- Link: <https://www.ratemyprofessors.com/professor/2098759>
- Professor Fahad Dogar's ratings (4.0/5.0 overall quality based on 5 ratings)

[CS112] Basic Course Information

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

- Course Title: Networks & Protocols (CS112)
- Instructor: Fahad Rafique Dogar
- Term: Fall 2024
- Total Enrollment: 20
- Survey Responses: 12
- Response Rate: 60.0%

[CS112] Fall 2025 Course Information

Source: Fall 2025 Course Catalog (<https://www.cs.tufts.edu/t/courses/description/fall2025/CS/112-01>)

1. Course Description:

Computer Networks and Protocols. Design and implementation of computer communication networks, protocols, and applications, with an emphasis on the Internet protocol suite. Network architectures and programming interfaces. Data link, transport, and routing protocols. Congestion sources and remedies. Addressing and naming in local area and wide area networks. Network security and network management.

2. Lecture Section:

- Section 01-LEC (Class No. 80418)
 - * Instructor: Fahad Rafique Dogar
 - * Schedule: Tu, Th 12:00PM – 1:15PM
 - * Location: Joyce Cummings Center, 170, Medford/Somerville
 - * Credits: 3
 - * Session: Regular
 - * Status: Open (green circle)
 - * Block: F+ Block

3. Meeting Dates: 09/02/2025 – 12/08/2025

4. Class Attributes:

- BFA-Science/Technology

- LA-Distribution-Mathematics
- SOE-Computing

5. Career: AS&E Undergrad/Grad

6. Enrollment Requirements:

- Prerequisite: Comp 15 and one of EE 14 or Comp 40
- Completion of CS 15 and either CS 40 or EE 14, or Graduate Standing

7. Enrollment Statistics:

- Enrollment: 70 capacity, 57 enrolled, 13 available seats
- Reserve Undergraduate Students: 60 capacity, 48 enrolled, 12 available
- Wait List: 5 capacity, 0 enrolled, 5 available

8. Additional Course Information:

- Textbook: Textbook info (link available in registration system)
- Grading: Graded
- Repeat for Credit: No
- Allow Multiple Enroll in Term: No
- Total Units Allowed: 3
- Total Completions Allowed: 1
- Instruction Mode: In Person

[CS112] RateMyProfessor Overview

Source: RateMyProfessor Reviews for CS112 (2016-2024)

- Reviews Timespan: January 2016 to December 2024
- Number of Reviews: 3
- Quality Rating Range: 4.0-5.0/5.0 (Average: 4.33/5.0)
- Difficulty Rating: Consistently 5.0/5.0 across all reviews
- Would Take Again: 66.7% Yes (2 Yes, 1 No)
- Most Common Grade: A
- Most Frequent Tags: "PARTICIPATION MATTERS", "GIVES GOOD FEEDBACK", "LOTS OF HOMEWORK"

[CS112] RateMyProfessor Quality Analysis

Source: RateMyProfessor Reviews for CS112 (2016-2024)

Rating Trends Over Time

- 2016 Rating: 4.0/5.0
- 2019 Rating: 4.0/5.0
- 2024 Rating: 5.0/5.0
- Analysis: Slight upward trend in quality ratings over the 8-year period

Instructor Strengths

Student Comments:

- "Networks is a fantastic class, and Fahad is a fantastic professor."

I've taken 8 CS electives at Tufts, and this was by far my favorite." (Dec 2024)

- "His lecture is good and I definitely have learned a lot of new knowledge from this course." (May 2019)
- "He is a really good lecturer." (Jan 2016)

Course Difficulty Assessment

Student Comments:

- "If you like systems and want to implement challenging assignments in C (like in 40), you will love this class." (Dec 2024)
- "The course load is really a hell. I need almost 20 hours per week to handle the assignment for getting a good grade. And the exam is very challengeable." (May 2019)
- "His exams are super tricky and it helps if you are a good notes taker." (Jan 2016)

[CS112] RateMyProfessor Course Workload Details

Source: RateMyProfessor Reviews for CS112 (2016-2024)

Time Commitment

Student Comments:

- "I need almost 20 hours per week to handle the assignment for getting a good grade." (May 2019)
- Consistent mention of high workload across all reviews
- All reviews rated difficulty as 5.0/5.0, indicating significant time investment required

Assignment Clarity

Student Comments:

- "His assignments instructions and expectations are not always clear (He has high expectations in everything) and you may do poorly as a result of that." (Jan 2016)

Grading Practices

Student Comments:

- "He is also a tough grader and gives A's to no more than 1/4 - 1/3 of the class." (Jan 2016)
- Despite tough grading, most reviewers reported receiving A grades

[CS112] RateMyProfessor Teaching Style

Source: RateMyProfessor Reviews for CS112 (2016-2024)

Key Teaching Characteristics

Student Comments:

- "If you go to every class and participate, you will love Fahad." (Dec 2024)
- Tagged as "AMAZING LECTURES" and "CARING" in recent review (Dec 2024)
- Tagged as "GIVES GOOD FEEDBACK" in multiple reviews

Attendance and Participation

- Attendance consistently marked as "Mandatory"
- "PARTICIPATION MATTERS" appears as a frequent tag
- Successful students emphasize the importance of attendance and active participation

[CS112] Overall Course Evaluation

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

- Overall Course Rating: 4.67/5.0 (Standard Deviation: 0.65)
- Computer Science Department Average Course Rating: 3.98/5.0 (Standard Deviation: 0.99)
- Course Rating Distribution:
 - * Excellent (5 points): 75.00% (9 students)
 - * Very Good (4 points): 16.67% (2 students)
 - * Satisfactory (3 points): 8.33% (1 student)
 - * Less Than Satisfactory (2 points): 0%
 - * Very Poor (1 point): 0%

[CS112] Instructor Evaluation

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

- Overall Instructor Rating: 4.58/5.0 (Standard Deviation: 0.90)
- Computer Science Department Average Instructor Rating: 4.16/5.0 (Standard Deviation: 0.99)
- Instructor Rating Distribution:
 - * Excellent (5 points): 75.00% (9 students)
 - * Very Good (4 points): 16.67% (2 students)
 - * Satisfactory (3 points): 0%
 - * Less Than Satisfactory (2 points): 8.33% (1 student)
 - * Very Poor (1 point): 0%

[CS112] Specific Course Evaluation Metrics

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

Course Objectives Achievement

- Average Score: 4.67/5.0 (Standard Deviation: 0.65)
- Excellent Rating Percentage: 75.00%
- Student Comment: "This course has really helped me turn the corner in terms of my confidence as an aspiring engineer. We've learned a ton of content and really gotten a chance to understand how the internet and computer networks work at a foundational level."

Classroom Time Utilization Efficiency

- Average Score: 4.75/5.0 (Standard Deviation: 0.45)
- Excellent Rating Percentage: 75.00%
- Student Comment: "Participation was highly encouraged, so hearing from my classmates often helped me understand the content better, and also helped build community within the classroom. The chalkboard style teaching was very engaging."

Effectiveness of Out-of-Class Activities (Readings, Assignments, Projects, etc.)

- Average Score: 4.67/5.0 (Standard Deviation: 0.49)
- Excellent Rating Percentage: 66.67%
- Student Comment: "The excellent projects. They were all a struggle but were incredibly satisfying to complete."

Student Interest Engagement

- Average Score: 4.67/5.0 (Standard Deviation: 0.65)
- Excellent Rating Percentage: 75.00%
- Student Comment: "This was the most challenging and fulfilling course I've ever taken."

[CS112] Instructor Teaching Ability Assessment

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

Classroom Organization Skills

- Average Score: 4.67/5.0 (Standard Deviation: 0.65)
- Excellent Rating Percentage: 75.00%
- Student Comment: "Smaller class size, more engaging class compared to bigger classes, the projects seem very useful."

Concept Explanation Ability

- Average Score: 4.67/5.0 (Standard Deviation: 0.49)
- Excellent Rating Percentage: 66.67%
- Student Comment: "Fahad is an outstanding instructor. He is very good at listening to students speak when they are confused, understanding the confusion, and addressing the specific gap in the student's understanding."

Feedback Timeliness

- Average Score: 4.42/5.0 (Standard Deviation: 0.79)
- Excellent Rating Percentage: 58.33%

Feedback Usefulness

- Average Score: 3.83/5.0 (Standard Deviation: 0.94)
- Excellent Rating Percentage: 25.00%
- Student Comment: "My only critique is that I think his feedback on assignments could be more detailed and more helpful."

Creating an Inclusive Classroom Environment

- Average Score: 4.67/5.0 (Standard Deviation: 0.65)
- Excellent Rating Percentage: 75.00%
- Student Comment: "It is more than evident that Fahad cares about his students beyond the classroom. He pushes for us to succeed and makes efforts to get to know us personally."

Out-of-Class Communication

- Average Score: 4.50/5.0 (Standard Deviation: 0.80)

- Excellent Rating Percentage: 66.67%
- Student Comment: "Engaging professor, he seems to enjoy engaging with people personally and getting to know them better."

[CS112] Course Impact Analysis

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

Depth of Learning Enhancement

Student Comments:

- "I was pushed by the professor to do my best, and his encouragement motivated me to study the content more rigorously."
- "This course has really helped me turn the corner in terms of my confidence as an aspiring engineer."
- "I've taken few courses (if any) that are as low-level and rigorous as Networks. Aside from learning about networking specifically, I believe the course enhanced my understanding of systems level programming."

Most Effective Learning Methods

Student Comments:

- "Participation was highly encouraged, so hearing from my classmates often helped me understand the content better, and also helped build community within the classroom."
- "The excellent projects. They were all a struggle but were incredibly satisfying to complete."
- "The projects were extremely helpful in cementing and solidifying the concepts learned in class."
- "Smaller class size, more engaging class compared to bigger classes, the projects seem very useful."

[CS112] Improvement Suggestions

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

Course Content and Structure

Student Suggestions:

- "Perhaps space out the research papers so they supported the content of the class right after we learned about a specific topic rather than having them all at the end."
- "I truly think every CS major would benefit from a version of this course with a lighter workload."
- "Make it WAYYYY more credits."
- "Fewer paper readings. I didn't really enjoy reading so many papers. Also, I wish the requirements for the final project were better defined."

Teaching and Support

Student Suggestions:

- "It would be lovely if students who couldn't make it to class were able to catch up with the material covered in the missed class somehow
- even though we don't have presentations we go through in class,

maybe a doc for each class with the basic outline could suffice."

- "Prof. Dogar's assignment specs do not give enough information to complete the assignment well. For example in the second assignment, the behavior of the chat program on error cases were to close connections immediately. This was not explicitly communicated through the spec but we only found out through going to lecture and asking."
- "I think that if Prof Dogar could just give us a simple client that didn't give us anything more than a simple sanity check, it would ease the burden of the work on students."

Course Policies

Student Suggestions:

- "I believe that Prof. Dogar means well for his students but can be unwavering in his course policies and inflexible to struggling students. I think his course policy on tokens can be an area of improvement because his current structure only allows students to submit one assignment late without any late penalties. I think having two one day tokens or charging students by the hour is a more reasonable approach because that allows the student to pick and choose when to exercise token usage."

[CS112] Instructor Personal Characteristics

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

Student Care

Student Comments:

- "It is more than evident that Fahad cares about his students beyond the classroom. He pushes for us to succeed and makes efforts to get to know us personally. This makes what we learned in class more memorable and the connection we have with him is something to be cherished years after we have taken this class."
- "He truly cares about his students – amazing professor."
- "Aside from the rigor and quality of the course content itself, I'm extremely appreciative of Professor Dogar. More than any professor I've had at Tufts, he was personable, encouraging, and respectful. He impressed upon us that while the coursework was challenging, we were all hard-working students that were capable of mastering the content."
- "It's obvious he cares about his students both in and outside the class – as evidenced both by the 1:1 meetings and the encouragement he gave (for example, by encouraging everyone to take care of their sleep)."
- "Thank you for a phenomenal semester Professor Dogar, I'm extremely grateful."

Teaching Ability

Student Comments:

- "He is clearly very knowledgeable of the subject he teaches and teaches it in a clear and engaging way."
- "Fahad is an outstanding instructor. He is very good at listening to students speak when they are confused, understanding the confusion,

and addressing the specific gap in the student's understanding."

[CS112] Physical Teaching Environment

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

Classrooms and Facilities

- Average Score: 3.75/5.0 (Standard Deviation: 0.87)
- Rating Distribution:
 - * Excellent: 25.00%
 - * Very Good: 25.00%
 - * Satisfactory: 50.00%
 - * Less Than Satisfactory: 0%
 - * Very Poor: 0%

[CS112] Course Difficulty and Workload

Source: Fall 2024 CS112 Networks & Protocols Course Evaluation Report

Course Challenge Level

Student Comments:

- "This was the most challenging and fulfilling course I've ever taken."
- "I've taken few courses (if any) that are as low-level and rigorous as Networks."
- "I truly think every CS major would benefit from a version of this course with a lighter workload. I am a fall semester senior, so I have been trying to take advantage of electives that are only offered in the Fall before I graduated, and taking this with other intense CS classes really made me struggle with sleep, anxiety, and mental health at times – especially during early midterm season where most projects were due."

Workload

Student Comments:

- "The projects were extremely helpful in cementing and solidifying the concepts learned in class."
- "It just seems that course is made intentionally harder by design and the Prof. has the power to make the work more reasonable for students but chooses not to."

[CS112] Fall 2024 Course Schedule, Policies, and Logistics

Source: Fall 2024 CS112 Course Syllabus

Course Logistics

- Lectures: MW 10:30–11:45 in Miner Hall, 224
- Staff:
 - * Fahad Dogar, Instructor (fahad@cs.tufts.edu)
 - Office Hours: Monday 3–4pm, JCC 440F
 - * Abdullah Bin Faisal, TA (Abdullah_Bin.Faisal@tufts.edu)
 - Office Hours: Scheduled around assignments
 - * Rukhshan Haroon, TA (Rukhshan.Haroon@tufts.edu)

- Office Hours: Scheduled around assignments

Course Materials and Grading

- Reference Book: Computer Networking: A Top Down Approach by Kurose and Ross (6th or more recent edition)
- Grade Distribution (tentative):
 - * Exams: 40%
 - * Assignments/Projects: 40%
 - * Class Participation: 20%

Tentative Schedule

- 9/4: Introduction (Chapter 1.1, 1.2, 1.3)
- 9/9: Introduction (Chapter 1.4 and 1.5)
- 9/11: Applications - Overview and Socket API (Chapter 2.1 and 2.7)
- 9/16: Applications - Web (Chapter 2.2)
- 9/18: Applications - DNS (Chapter 2.5)
- 9/23: Applications - P2P (Chapter 2.6)
- 9/25: Transport - Overview + UDP (Chapter 3.1, 3.2, and 3.3)
- 9/30: Transport - Reliability (Chapter 3.4)
- 10/2: Transport - TCP (Chapter 3.5)
- 10/7: Transport - Congestion Control (Chapter 3.6 and 3.7)
- 10/09: Network Layer - IP (Chapter 4.1, 4.2, 4.3, 4.4)
- 10/14: No Class
- 10/16: Routing - BGP (Chapter 4.6) and OSPF (Chapter 4.5)
- 10/21: Link Layer (Chapter 5.1, 5.2, 5.3, 5.4)
- 10/23: PHY + wireless (Chapter 6.1 and 6.3)
- 10/28: Midterm
- 10/30: Midterm Review + Wireless Wrap up
- 11/4: Security (Chapter 8.1 - 8.4; 8.6)
- 11/6: Technology discussion (QUIC/HTTP 3.0); Paper: Improving Battery life (Catnap)
- 11/12 (Tuesday): Tech Discussion: CDN/Akamai; Paper: Missed Calls for Cheap Communication in Developing Countries
- 11/13: LLM Proxy - paper
- 11/18: Tech Discussion: Software-defined Networking; Paper: Making the most of the broadcast nature of wireless (Network Coding)
- 11/20: DoS/DDoS; Paper1: ddospaper.pdf; Paper 2
- 11/25: TCP in the modern world
- 12/2: TCP inside data centers. DCTCP
- 12/4: Future Internet Architecture
- 12/9: Project Demos + Review/Feedback
- Final Exam: Tuesday December 17th 3:30-5:30pm (per block schedule)

Course Policies

- Use of cell phones and other electronic devices isn't allowed in the class without prior approval of the instructor.
- Class attendance is mandatory. Please aim to arrive 5 minutes early so we can start on time. If you absolutely have to miss a class, please inform the instructor in writing.
- Class participation grade includes attendance, in-class activities,

participation on Piazza, asking questions during class, etc.

- Programming assignments can be submitted up to two days late with a penalty of 10% per extra day. You are allowed one late submission without grade penalty (but it should still be within two days of the submission due date).
- Collaboration on programming assignments is encouraged but you should submit your own work (see the university policies on plagiarism).
- You shouldn't share the course material (lecture videos, content, assignments/exams) with anyone who is currently not enrolled in the class (if you feel there is a good reason to do so, please get instructor approval first).
- Use of AI tools (e.g., chatGPT) isn't allowed unless explicitly stated otherwise.

Student Advice and Inclusion

- Please get a good night's sleep (not just before this class but everyday) -- it will drastically improve your learning in class.
- Try to engage with your peers -- make friends while you learn about networks!
- Everyone should feel welcomed and included.
- If for any reason you feel less included in the class because of your background, please reach out to the instructor or the diversity coordinator of the CS department.
- If you have a disability that requires reasonable accommodations, please contact the Student Accessibility Services office at Accessibility@tufts.edu or 617-627-4539 to make an appointment with an SAS representative to determine appropriate accommodations. Please be aware that accommodations cannot be enacted retroactively, making timeliness a critical aspect for their provision.

Last Updated: Based on Fall 2024 Evaluation and RateMyProfessor Reviews (2016-2024)

Data updated on April 20, 2025.